



Original Article

Metabonomics and the Gut Microbiome Associated With Primary Response to Anti-TNF Therapy in Crohn's Disease

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Abstract

Background and Aims: Anti-tumour necrosis factor [anti-TNF] therapy is indicated for treatment of moderate to severe inflammatory bowel disease [IBD], but has a primary non-response rate of around 30%. We aim to use metabonomic and metataxonomic profiling to identify predictive biomarkers of anti-TNF response in Crohn's disease.

Methods: Patients with luminal Crohn's disease, commencing anti-TNF therapy, were recruited with urine, faeces, and serum samples being collected at baseline and 3-monthly. Primary response was defined according to a combination of clinical and objective markers of inflammation. Samples were measured using three UPLC-MS assays: lipid, bile acid, and Hydrophilic Interaction Liquid Chromatography [HILIC] profiling with 16S rRNA gene sequencing of faeces.

Results: Samples were collected from 76 Crohn's disease patients who were anti-TNF naïve and from 13 healthy controls. There were 11 responders, 37 non-responders, and 28 partial responders in anti-TNF-treated Crohn's patients. Histidine and cysteine were identified as biomarkers of response from polar metabolite profiling [HILIC] of serum and urine. Lipid profiling of serum and faeces found phosphocholines, ceramides, sphingomyelins, and triglycerides, and bile acid profiling identified primary bile acids to be associated with non-response to anti-TNF therapy, with higher levels of phase 2 conjugates in non-responders. Receiver operating curves for treatment response demonstrated 0.94 $\pm$ 0.10 [faecal lipid], 0.81 $\pm$ 0.17 [faecal bile acid], and 0.74 $\pm$ 0.15 [serum bile acid] predictive ability for anti-TNF response in Crohn's disease.

Conclusions: This prospective, longitudinal cohort study of metabonomic and 16S rRNA gene sequencing analysis demonstrates that a range of metabolic biomarkers involving lipid, bile acid, and amino acid pathways may contribute to prediction of response to anti-TNF therapy in Crohn's disease.

Podcast: This article has an associated podcast which can be accessed at <https://academic.oup.com/ecco-jcc/pages/podcast>

Key Words: Crohn's disease; metabolomics; microbiota

1. Introduction

Inflammatory bowel diseases [IBD], which include Crohn's disease [CD] and ulcerative colitis [UC], are lifelong and life-limiting, relapsing and remitting inflammatory disorders that affect the intestine.¹ The aetiology is poorly defined, but it is clear that both environmental and genetic factors predispose an individual to the development of IBD.^{2,3} With regard to the contribution of genomics to the field of IBD, over 250 host susceptibility loci have been identified to date.^{4–7} It is evident that in addition to these loci, there are environmental triggers and other host factors that play an important role in pathogenesis and in determining specific phenotypes which predict response to IBD therapies.⁶

There are several new therapeutic targets for IBD,^{8,9} with recent therapeutic strategies demonstrating that early initiation of treatment results in greater response.¹⁰ Personalising treatments to select the optimal therapy for disease control is crucial in IBD, as the likelihood of response may decrease with each sequential biologic agent.⁹ Anti-tumour necrosis- α [TNF] remains the backbone of treatment for patients with moderate to severe CD, with primary response rates of 60–70% in initial controlled trials.¹¹ Currently, there is interest in biomarkers for predicting response to anti-TNF therapy, including using clinical factors such as age, duration of disease, or body mass index [BMI].¹² There are promising biomarkers which have been investigated but are not yet available in clinical practice, including oncostatin M, IL13RA2, and TREM-1.^{13–15} For other biologic drugs such as etrolizumab, an α E β 7 anti-integrin, there is the suggestion that colonic expression of α EA is predictive of response.¹⁶ Recently, in a prospective cohort of IBD patients [42 CD and 43 UC] starting therapy with vedolizumab, an α 4 β 7 anti-integrin, the composition of the microbiota in the colonic microbiome at baseline was used to predict response to vedolizumab. The researchers demonstrated that *Roseburia inulinivorans* and a *Burkholderiales* sp. were more abundant at baseline among those CD patients achieving Week 14 remission.¹⁷ Additionally, community α -diversity at baseline was significantly higher in CD patients who achieved remission, but β -diversity was lower at baseline among remitters. Changes in bacterial function were associated with response to biologic therapy. Specifically, 13 pathways, including branched-chain amino acid synthesis, were enriched in CD patients achieving remission on vedolizumab.

The microbiota is involved in driving inflammation in IBD,¹⁸ and a dysregulation of the microbiome has been reported in both CD and UC, with reduced diversity and temporal instability in the dominant taxa compared with healthy individuals.¹⁹ For example, reduced relative abundance of *Firmicutes* and anti-inflammatory micro-organisms [*Faecalibacterium prausnitzii*, *Bifidobacterium adolescentis*, and *Dialister invisus*] have been demonstrated in analysis of faecal microbiota from CD patients.^{18,20–22} Microbiota sampling, using a longitudinal approach, has been shown to partially overcome the effects of microbial instability and more accurately classify CD patients based on their microbiota signature.²³ Metabolic profiling of patients with IBD has also indicated that medium-chain fatty acid, such as pentanoate, hexanoate, heptanoate, octanoate, and nonanoate, are directly associated with disease activity in IBD.^{19,24,25}

Metabolic phenotyping by high-resolution spectroscopy, either liquid chromatography-mass spectrometry [LC-MS] or nuclear magnetic resonance spectroscopy [NMR], provides an additional tool for establishing the functionality of the gut microbiome and allows clinicians and scientists to gain new insights into the knowledge gap which exists in assessing the enteric microbial composition and their function. This metabonomic approach allows the detection of

metabolites in IBD upon which hypotheses about their role in disease activity or health can be generated. Gut microbiota compositional differences in IBD patients has been linked to changes in metabolite concentrations and diversity such as bile acids [BA], lipids [including fatty acids], amino acids, and eicosanoids, all of which are important in inflammation, energy, and signalling.^{24,26,27} For example, metabolic profiling of bile acid metabolism in IBD results in alteration in gut barrier homeostasis and hence intestinal inflammation.^{19,28}

Using a prospectively recruited cohort of patients with Crohn's disease before commencing anti-TNF therapy, faecal, urine, and serum samples from IBD patients were used to identify metabonomic and microbiota predictors of response to anti-TNF treatment. Although the emphasis was on CD, samples were also collected from patients with moderate to severe ulcerative colitis, to serve as a disease control since previous work has demonstrated distinct differences in metabonomic profiles. We classified patients' response according to clinical indices and objective markers of inflammation with C-reactive protein [CRP], faecal calprotectin [FC], and endoscopy/ imaging performed as part of routine care.

2. Methods

2.1. Sample collection

Serum, urine, and faecal samples were obtained from three cohorts: patients with CD [$n = 76$], UC [$n = 10$] [Figure 1], and healthy participants [HC, $n = 13$]. Patients with CD had induction doses of anti-TNF therapy starting at Day 0 with either adalimumab [Humira®] or infliximab [Remicade®] [Figure 1A]. Patients with UC were inpatients requiring salvage therapy for severe acute colitis without previous anti-TNF exposure and served as a comparison within our IBD cohort. An aliquot of samples with urine [5 mL], faeces [2 g], and serum [6 mL] were immediately frozen at -20°C and transported within 24 h to a -80°C freezer for storage before analysis. Ethical approval was obtained from Brent NRES Committee London with HREC 08/H0717/24.

2.2. Baseline disease activity assessment

Participants in the control group had no previous history of gastrointestinal or metabolic disease and had not used antibiotics or undergone medical therapy influencing gut transit or microbiota in the preceding 6 months. Diagnosis of IBD [CD and UC] was confirmed by endoscopy, histology, and radiological techniques. In addition, disease activity was assessed for the CD group applying clinical indices with CD Activity Index [CDAI] scores and, for the UC cohort, the Mayo score.^{29,30} All patients had an objective measure of disease activity with FC and CRP at baseline. Exclusion criteria included any patient with a previous history of surgical resection of the gastrointestinal tract. Participants who met the control and IBD group criteria were included in this study and provided written informed consent. Clinical and biochemical characteristics of IBD patients are summarised in Table 1.

Patient's response was determined according to clinical indices and objective markers including FC, CRP, and radiological or endoscopic improvement [Supplementary Table S1, available as Supplementary data at ECCO-JCC online.]

2.2.1. Chemicals and reagents for metabonomic profiling assays

Organic solvents [HPLC grade] used for sample preparation were obtained from Sigma-Aldrich [Dorset, UK]. Mobile phases were prepared with grade LC-MS solvents and modifiers from Sigma-Aldrich [Dorset, UK].

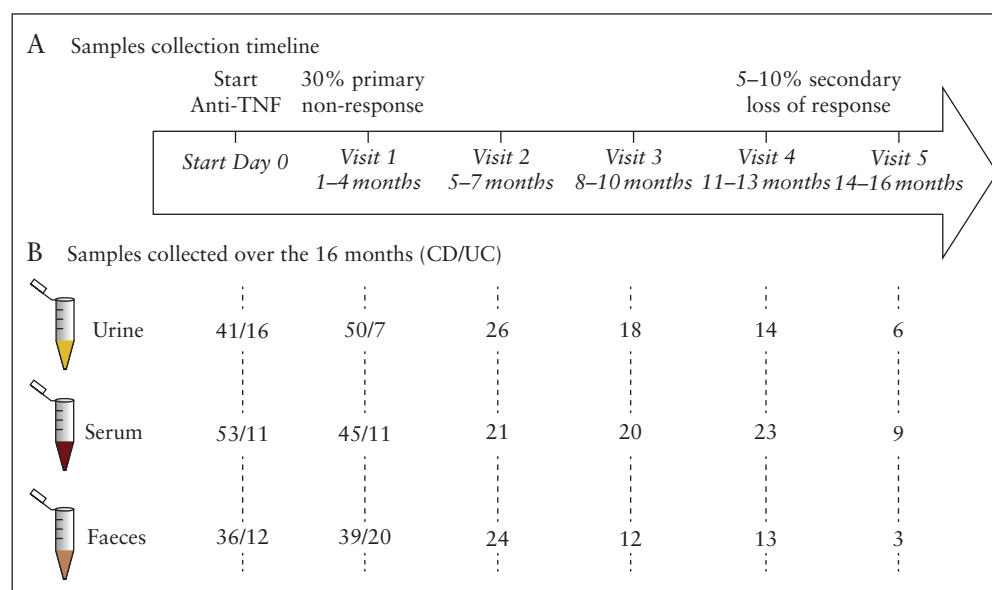


Figure 1. Longitudinal study on IBD patients' response to anti-TNF treatment [A]. Urine [$n = 181$], serum [$n = 214$], and faecal [$n = 147$] samples were collected from CD and UC, with patient samples over 16 months [B]. IBD, inflammatory bowel disease; CD, Crohn's disease; UC, ulcerative colitis.

Table 1. Summary of clinical and biochemical characteristics of IBD patients.

		Normal range	CD	UC	<i>p</i> -value
Inflammatory biomarkers	Infliximab/adalimumab		66/10	10	
	Age [IQR]	30.5 2.94 [27–35] ^a	38.49 14.61 [19–82]	35.52 17.34 [19–70]	0.45
	% female	54% ^a	53%	48%	0.56
	Haemoglobin [g/L] [IQR]	120–175	127.71 14.52 [87–159]	123.06 17.91 [98–152]	0.37
	White cell count [$\times 10^9/L$] [IQR]	4.5–10	7.51 2.92 [2.9–20.1]	10.3 4.35 [4.4–18.1]	0.03
	Platelet count [$\times 10^9/L$] [IQR]	150–450	326.23 94.74 [113–561]	352.33 83.78 [244–544]	0.31
	ESR [mm/h] [IQR]	0–29	17.28 18.15 [1–95]	11 9.91 [2–21]	0.22
	C-reactive protein [mg/L] [IQR]	< 5	12.49 14.85 [1–78]	7.35 6.46 [1–17]	0.06
	Calprotectin [$\mu g/g$] [IQR]	0–250	699.22 655.78 [7–2000]	534.21 446.9 [5.3–1170]	0.34
	Bilirubin [μM] [IQR]	3–25	7.55 8.22 [2–59]	6 2.85 [3–14]	0.25
Liver function	Alkaline phosphatase [IU/L] [IQR]	44–147	83.89 29.79 [41–190]	80.5 26.52 [48–156]	0.68
	Alanine aminotransferase [IU/L] [IQR]	20–60	25.56 15.26 [7–73]	31.35 33.34 [11–114]	0.53
Renal function	Albumin [g/L] [IQR]	35–55	42.58 4.23 [28–55]	40.64 6.89 [28–49]	0.33
	Creatinine [μM] [IQR]	45–110	69.12 16.62 [34–113]	61.21 13.65 [35–86]	0.08
	CDAI/Mayo index [IQR]	-	490 [409–501]	12 [10–12]	-

^aFigures represent age and sex of healthy controls.

IBD, inflammatory bowel disease; CD, Crohn's disease; UC, ulcerative colitis; IQR, interquartile range; ESR, erythrocyte sedimentation rate; CDAI, Crohn's Disease Activity Index.

2.2.2.2. Metataxonomics [16S rRNA gene sequencing]

Sequencing of the 16S rRNA gene was implemented to characterise the bacterial composition of the faeces. DNA was extracted from 250 mg of faeces using the PowerLyzer PowerSoil DNA Isolation Kit [Mo Bio, Carlsbad, CA, USA] following manufacturer's instructions with the

following modification: samples were homogenised in a Bullet Blender Storm bead beater [Chembio Ltd, St Albans, UK] for 3 min at speed 8 to aid in bacterial cell lysis. DNA was aliquoted and stored at $-80^{\circ}C$ until ready to use. Sample libraries were prepared following Illumina's 16S Metagenomic Sequencing Library Preparation Protocol.³¹ The

SequalPrep Normalization Plate Kit [Life Technologies, Paisley, UK] was used to purify and normalise the sample libraries following the index polymerase chain reaction [PCR] amplification step. Also, the NEBNext Library Quant Kit for Illumina [New England Biolabs, Hitchin, UK] was used to quantify the sample libraries. Pooled sample libraries were sequenced using the Illumina MiSeq platform [Illumina Inc., Saffron Walden, UK] and the MiSeq Reagent Kit v3 [Illumina] using paired-end 300-bp chemistry.

16S rRNA gene sequencing data was pre-processed using the Mothur package version v.1.35.0 following the MiSeq SOP Pipeline.³² Sequence alignments were performed using the Silva bacterial database [www.arb-silva.de/] and sequences were classified using the Ribosomal Database Project [RDP] database reference sequence files and the Wang method.³³ Comparison of pre- vs post-anti-TNF therapy samples [taken 5–7 months after starting therapy] was performed using the Statistical Analysis of Metagenomic Profiles [STAMP] software package with White's non-parametric t test and Benjamini-Hochberg correction to adjust for false-discovery rate.³⁴

2.2.3. Metabolic profiling of serum, urine, and faeces

Three analytical platforms were implemented to achieve broad metabolome coverage: polar metabolites [hydrophilic interaction liquid chromatography, HILIC], lipids, and bile acids ultra-performance LC [UPLC]-MS profiling, combined with 16S rRNA gene sequencing. A comprehensive analysis of metabolites was performed on serum, urine, and faecal samples from the 112 participants [Figure 1] using UPLC-MS profiling analyses.

2.2.4. Sample preparation for metabolic profiling of serum

Before analysis, lipids/polar metabolites and bile acids were extracted, respectively, with pre-chilled isopropanol and methanol from 50 μ L serum [1 aliquot/3 solvent].³⁵ Aliquots were vortexed and centrifuged at 14 000 g for 20 min at -4°C , and 20 μ L of supernatant was transferred into vials for the lipid and bile acid assay. The supernatant for the polar metabolites assay was evaporated at room temperature in a vacuum concentrator [Eppendorf Concentrator Plus], reconstituted in acetonitrile/ H_2O [90:10], and transferred into vials. In addition, quality control [QC] samples were prepared from pooled serum samples [20 μ L from each sample]. Samples were stored at -80°C awaiting UPLC-MS analysis.

2.2.5. Sample preparation for urine

Aliquots [200 μ L] were evaporated and reconstituted with H_2O /isopropanol/acetonitrile [2:1:1] for the bile acid assay and H_2O /acetonitrile [10:90] for the HILIC polar metabolite assay. Samples were vortexed and centrifuged at 14 000 g for 20 min at -4°C . Sample supernatants were transferred into vials, and the QC samples were prepared from pooled urine samples [20 μ L from each sample]. Samples were stored at -80°C awaiting UPLC-MS analysis.

2.2.6. Sample preparation for faeces

Before analysis, faecal samples were weighed [≈ 150 mg] and dried. For the lipid, bile acid, and polar metabolite profiling assays, samples were extracted with mixture solvents of H_2O /isopropanol/acetonitrile [respectively 1:1:2, 2:1:1, and 1:1:1]. Samples were vortexed and centrifuged at 14 000 g for 20 min at -4°C . The supernatant of samples prepared for the polar metabolite assay was evaporated and reconstituted with acetonitrile/ H_2O [90:10]. For each sample, the supernatant fraction was transferred into a vial and a QC sample was prepared from pooled faeces samples [20 μ L from each sample]. Samples were stored at -80°C awaiting UPLC-MS analysis.

2.2.7. Metabolic profiling assays

Aqueous and organic extractions were performed for serum, urine, and faecal samples. All samples were transferred into vials with QC samples prepared from pooled faecal samples [20 μ L from each sample]. These samples were analysed using reversed phase UPLC-MS lipid profiling, HILIC UPLC-MS profiling, and bile acid UPLC-MS profiling.³⁵ The QC samples were injected every 12 samples throughout the analytical run. Samples were stored at -80°C awaiting UPLC-MS analysis.

2.2.8. Data pre-processing and statistical analysis of MS data

The UPLC-MS raw data were converted to netCDF format using MassLynxTM software [Waters Corporation, Milford, USA]. Data processing was implemented using the XCMS package in R programming software. The dataset was normalised to total spectral area. Multivariate data analysis was performed using the SIMCA package [v.13.0.2, Umetrics, Umeå, Sweden]. Multivariate Pareto scaled data were modelled using principal components analysis [PCA] and orthogonal partial least squares discriminant analysis [OPLS-DA]. The OPLS-DA parameters [R^2X , R^2Y , Q^2Y , and cross-validated CV ANOVA p -value] aimed to assess the inherent metabolic variation and robustness of each model. The p -value was adjusted for multiple testing. The two-tailed t test was applied to the main discriminatory features of each model [Office Excel 2007]. A Python script was developed to plot the receiver operating characteristic [ROC] and calculate the area under the curve [AUC] with confusion matrix parameters [accuracy, sensitivity, and specificity] of each biomarker.

2.2.9. Metabolite assignment

Metabolite identification by MS was conducted by matching accurate m/z measurements of detected chromatographic peaks to theoretical values from in-house databases and online databases such as the human metabolite database HMDB, [<http://www.hmdb.ca/>], KEGG [<http://www.genome.jp/kegg/ligand.html>], LIPID MAPS [<http://www.lipidmaps.org/tools/index.html>], and METLIN [<http://metlin.scripps.edu/>]. Tandem MS fragmentation patterns were obtained for further structural elucidation and confirmed with an authentic standards matching for retention time and m/z .

3. Results

3.1. Demographics

A total of 86 IBD patients were included in this study [76 CD, 10 UC]. All were anti-TNF naïve. The median age was 38 years for CD and 35 years for UC patients. There were no significant differences in demographics of the IBD patients compared with controls [Table 1]. CD patients had a baseline CDAI of 490 (interquartile range [IQR] [409–501]) with a median CRP of 12.5 mg/L and FC of 699 $\mu\text{g/g}$. UC patients had a baseline Mayo score of 12 [10–12], CRP of 7.3 mg/L, and FC of 534 $\mu\text{g/g}$. At Week 102, 56% of Crohn's patients remained in remission according to objective markers of disease activity [CRP and FC]. A total of 37 CD participants were classified as primary non-responders to anti-TNF therapy, with objective markers previously described with 11 responders and 28 partial responders in the CD cohort. All patients who were classified as responders were also responders according to CDAI criteria, with a decrease in CDAI of 100 points.

Due to the existing literature demonstrating clear differences in metabolic profiling between CD and UC, we performed separate analyses on these cohorts. However, due to the small sample size of UC patients, analysis was only performed to demonstrate differences

at baseline between the two groups. Predictive metabolic profiling analysis for primary non-response was only undertaken for CD patients.

Differences in metabolic profiles were observed between healthy controls, CD patients, and UC patients, using UPLC-MS metabolomic approaches

Robust OPLS-DA models, as assessed by the QC sample distribution and model validation statistics [Q^2Y , permutation testing],³⁶ were built from lipid, polar metabolites, and bile acid profiles and demonstrated significant metabolic differences between the three groups: healthy controls, CD, and UC [Supplementary Figures S1–3] and [Table S2, available as Supplementary data at ECCO-JCC online].

3.2. Differences in metataxonomic profiles observed between healthy controls and CD patients

Profound changes in gut microbiota composition between HC, UC, and CD patients were identified from the metataxonomic analysis of faecal samples [Figure 2].

The gut microbiota from CD patients was characterised by higher relative abundance of the phylum Firmicutes and a lower relative abundance of the phylum Bacteroidetes compared with healthy controls [Figure 2]. At the family level, there was a significant decrease in the relative proportion of the families Bacteroidaceae, Ruminococcaceae, unclassified Clostriales, and Rikenellaceae, combined with higher relative proportions of the family Lachnospiraceae and Lactobacillaceae as observed in CD patients. There was also a significant reduction in the relative abundances of the genera Bacteroides, Faecalibacterium, unclassified Clostriales, and Alistipes, as well as an increase in the relative proportion of unclassified Lachnospiraceae and Lactobacillus in CD patients compared with healthy controls [Figure 2].

3.3. Metabolic predictors of anti-TNF response using UPLC-MS profiling in CD patients.

OPLS-DA was carried out to determine if there was a metabolic response to the anti-TNF treatment of CD patients. Using clinical and objective markers of disease activity, OPLS-DA models were built from metabolic markers associated with response to anti-TNF therapy. The subsequent identified metabolites were found to be biomarkers associated with anti-TNF response.

Classifying the response in CD patients to anti-TNF treatment, according to clinical and objective markers, showed good correlation with metabolic profiles [Supplementary Table S4, Figures S4–7, available as Supplementary data at ECCO-JCC online]. The OPLS-DA models built from all the visits were validated with p -value <0.05 across all UPLC-MS assays and biofluids. Based on baseline samples in comparison with later time points [visit 4 to visit 6, which were at 11 to 16 months], OPLS-DA models were significant for bile acid profiling in serum and urine, HILIC profiling in serum, and lipid profiling in faeces for responders and non-responders

Bile acid metabolism was investigated by UPLC-MS profiling. Several compounds from faeces [$n = 5$ bile acid markers], serum [$n = 5$ bile acid markers], and urine [$n = 3$ bile acid markers] differentiated between patients responding and not responding to anti-TNF treatment [Supplementary Table S5 and Figures S4–7, available as Supplementary data at ECCO-JCC online]. Bile acid markers associated with the sulphate/taurine, glycine, and sulphate/glycine pathway respectively, were altered in faeces [Figure 3A], serum [Figure 3B], and urine [Figure 3C]. Although the bile acid

profiles from all three biofluids discriminated between responders and non-responders, each biofluid manifested a distinct bile acid signature [Figure 3 and Table S5]. Secondary [deoxycholic acid] and tertiary bile acid markers were significantly elevated and highly represented in serum of patients responding to anti-TNF treatment. Patients who did not respond to the anti-TNF treatment exhibited significantly higher levels of sulphate and glycine-conjugated primary bile acids. From a combination of the three faecal bile acid markers with the highest discriminating ability, an AUROC of 0.81 ± 0.17 was generated, demonstrating a model with good predictive ability for anti-TNF treatment response [Figure 4C]: five serum bile acid markers produced a model with AUC 0.74 ± 0.15 [Figure 4B] and urine BA of 0.70 ± 0.17 [Figure 4E]. Combining the three biofluid matrices did not substantially improve the prediction of response.

The polar metabolite [HILIC] profiling of serum, urine, and faeces indicated significant differences between responders and non-responders to anti-TNF therapy [Supplementary Table S4 and Figure S6, available as Supplementary data at ECCO-JCC online]. Histidine levels in faeces and serum were identified to be significantly higher in anti-TNF responders [Figure 5A, B], together with a number of unidentified metabolites. Urinary cysteine was also elevated in responders [Figure 5C]. ROC analysis demonstrated a good model for predicting anti-TNF response for urinary cysteine [AUC 0.7 ± 0.14] [Figure 4H] but not for histidine in faeces [AUC = 0.63 ± 0.28] [Figure 4F] or histidine from serum [AUC = 0.48 ± 0.18] [Figure 4G].

Metabolic variations between responders and non-responders to anti-TNF treatment were also observed in the lipid profiles [Table S4 and Figures S4–7, available as Supplementary data at ECCO-JCC online]. Circulatory lipid markers, phosphatidylcholine [PC], ceramides [Cer], sphingomyelins [SM], and triglycerides [TG], were present in significantly lower concentrations in patients who responded to the treatment [Figure 6B]. Conversely, increased levels of phosphocholines and triglycerides were detected in the faeces of responders [Figure 6A]. ROC analysis demonstrated a good model for predicting anti-TNF response for serum lipid [AUC 0.78 ± 0.12] [Figure 4A] and a very high predictive accuracy for faecal lipid profiling [AUC = 0.94 ± 0.10] [Figure 4B]. The likelihood ratios for each AUC are shown in Table 2.

3.4. Exploration of 16S rRNA gene sequencing of longitudinal samples with respect to response to anti-TNF therapy

An analysis of all samples from IBD patients before anti-TNF therapy compared with all IBD patients after anti-TNF therapy [including responders, non-responders, and partial responders] showed that there was a decrease in the phylum Bacteroidetes [$p = 0.020$], corresponding to the class Bacteroidia [$p = 0.038$] following anti-TNF therapy [Figure 7]. However, we did not find any differences in the gut microbiota profiles of responders compared with non-responders, either before or after anti-TNF therapy. This is most likely due to sample size limitations.

4. Discussion

Anti-TNF therapy primary non-response in patients with Crohn's disease remains an issue with rates of 30% in the reported literature.³⁷ Biomarkers for the prediction of anti-TNF therapy response have hitherto been lacking.¹¹ We identified metabolic profiles, which are predictive of primary non-response in Crohn's patients

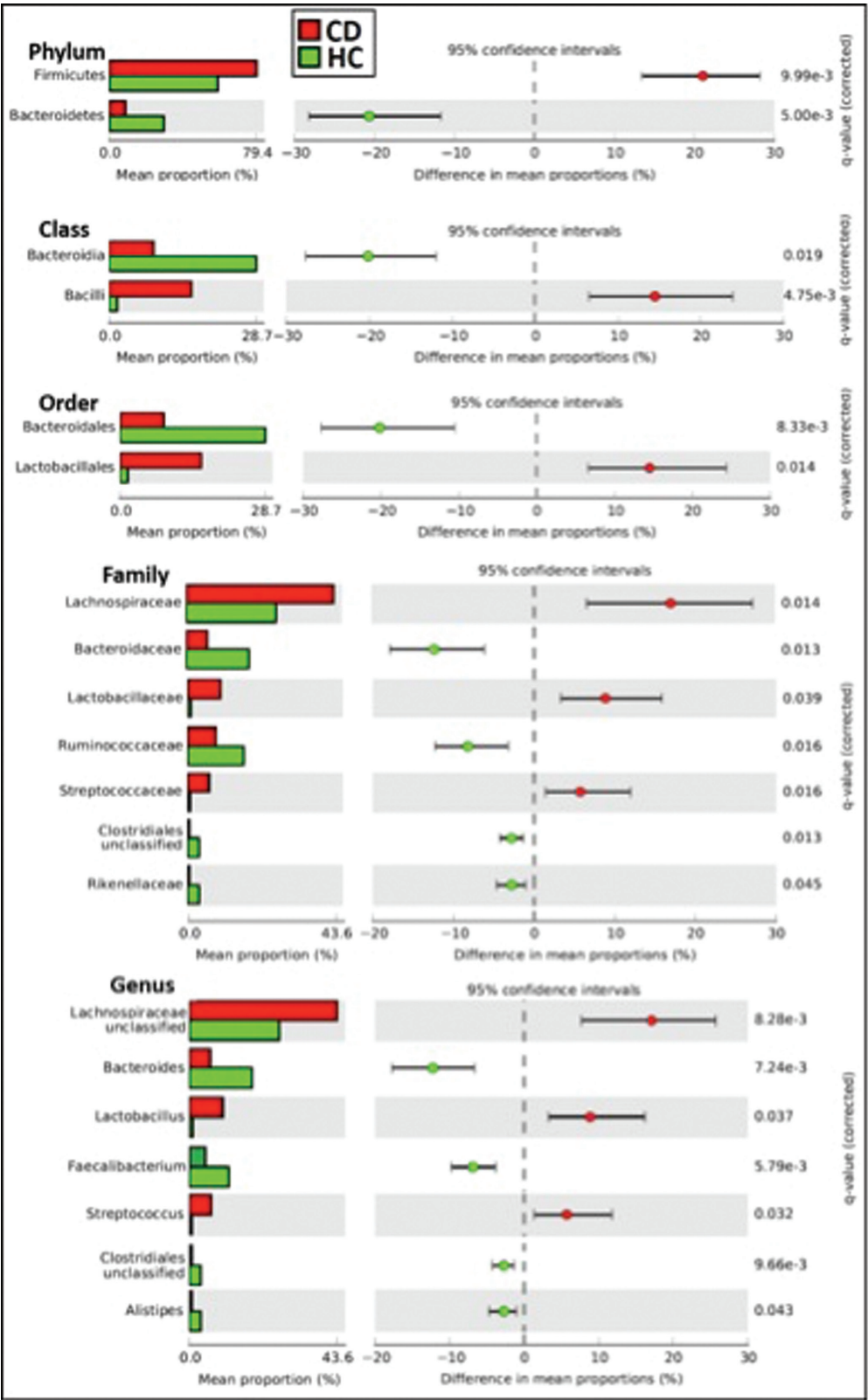


Figure 2. Significant variations in faecal bacterial composition of healthy control [HC] and Crohn's disease [CD] patients observed at all taxonomic levels. Faecal samples were analysed by 16S rRNA gene sequencing, and data were evaluated using White's non-parametric t test with Benjamini-Hochberg false-discovery rate [FDR].

undergoing anti-TNF therapy, with alterations in bile acids and amino acid and lipid metabolites. In addition to identifying a serum metabolic phenotype associated with anti-TNF response, we

demonstrated that we were also able to classify responders to anti-TNF therapy with urinary cysteine and bile acid-specific profiles in serum, urine, and faeces.

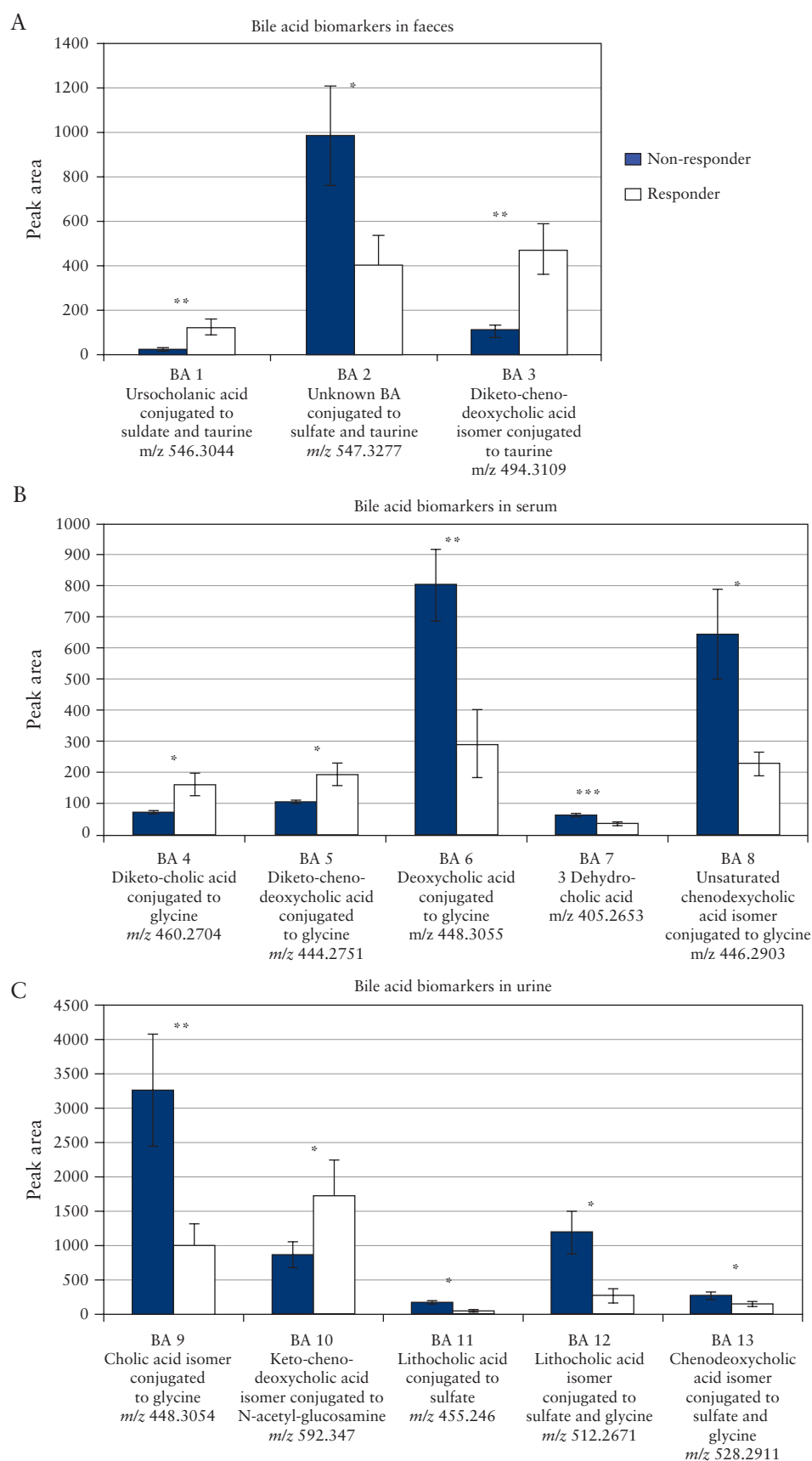


Figure 3. BA metabolites in urine, serum and faeces correlate with the longitudinal response of CD patients to anti-TNF treatment. BA markers vary between responders and non-responders to anti-TNF therapy in faeces [A], serum [B], and urine samples [C]. Two-tailed Student's *t* test, **p* < 0.05, ***p* < 0.01, and ****p* < 0.001. Supplementary information is available in Supplementary Table S5, available as Supplementary data at ECCO-JCC online. CD, Crohn's disease; BA, bile acids.

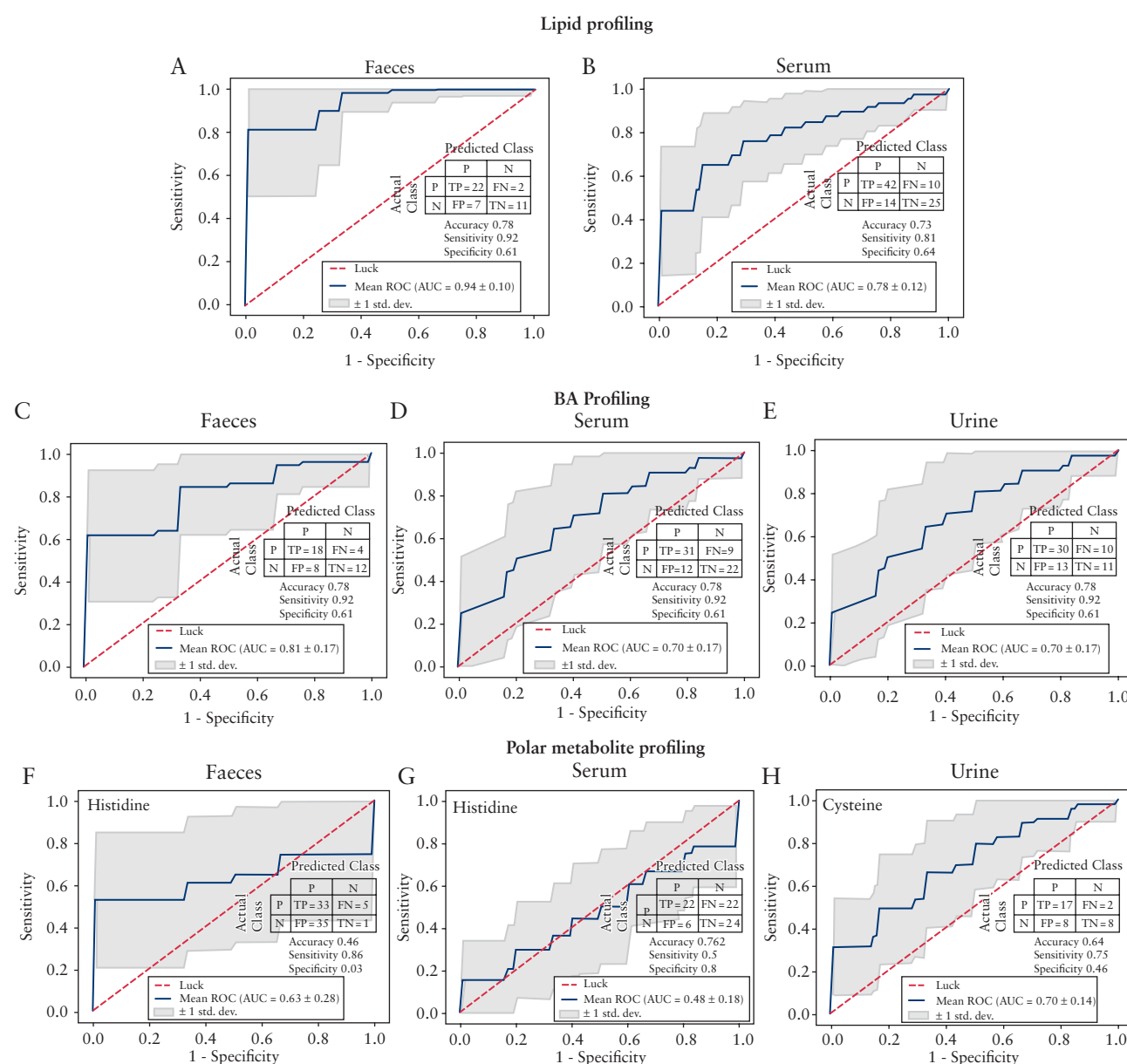


Figure 4. Predictive model demonstrating area under curve for classifying non-responders to anti-TNF therapy from lipid, bile acid, and HILIC profiling biomarkers. [A] Serum lipid. [B] Faecal lipid. [C] Faecal bile acid. [D] Serum bile acid. [E] Urine bile acid. [F] Faecal histidine. [G] Serum histidine. [H] Urinary cysteine: biomarkers for predicting non-response to anti-TNF therapy. Confusion matrix parameters are true positive [TP], true negative [TN], false negative [FN], and false positive [FP] with calculated accuracy, specificity, and sensitivity.

4.1. IBD biofluid metabolite and metataxonomic profiles exhibit a metabolic phenotype distinct from that of healthy controls

Previous literature in the field of metabonomics has seen the identification of specific biochemical profiles that separate IBD patients from healthy controls.²⁷ Most studies to date comparing CD and UC have been challenging, largely due to small patient numbers and ill-defined cohorts.³⁸ A recent publication by Kiel *et al.* identifies metabolite function alterations in anti-TNF remitters compared with non-remitters in a population of IBD and rheumatoid arthritis patients.³⁹ Our study applies lipid, bile acid, and polar metabolite [HILIC] profiling [serum, urine, and faecal samples] to a cohort of 76 CD patients and 10 UC patients profiled over a

16-month period, and demonstrates that severe UC can be separated distinctly from moderate to severe CD, confirming the work of Scoville *et al.* who showed differences between the two disease states. The common core of metabolites shared by UC and CD which differentiate healthy from IBD was found to account for 35% of the metabolites measured, which was similar to the percentage of IBD dysregulated metabolites identified by Scoville *et al.* [31%].⁴⁰ IBD has been associated with increased serum concentrations of primary bile acids, plasmalogens, and lysolipids, and tricarboxylic acid cycle [TCA] intermediates, sterols, sphingolipids, acylcarnitines, and endocannabinoids were found in lower concentrations in the serum of IBD patients.^{41,42} We also found higher concentrations of serum bile acids in the IBD patients. However, in contrast we found lower

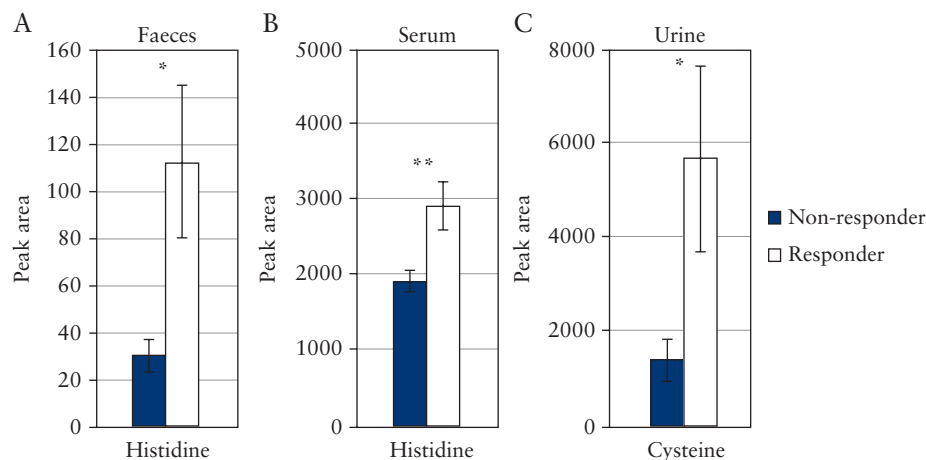


Figure 5. Polar metabolites [HILIC] correlating with the longitudinal response of CD patient's anti-TNF treatment. Histidine and cysteine vary between patients who did or did not respond to anti-TNF treatment in faeces [A], serum [B], and urine samples [C]. Two-tailed Student's *t* test, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. CD, Crohn's disease.

serum lysophospholipid concentrations in IBD patients and higher levels of sphingolipids, acylcarnitines, sterols, and TCA intermediates. Increased serum acylcarnitines have been reported in canine IBD.⁴³ The difference in cohort composition may account for some of these discrepancies, since the Scoville study assessed 20 CD and 20 UC patients whereas the current study was intentionally biased towards CD [$n = 76$] with a smaller UC group [$n = 10$].

Based on baseline samples, only faecal and urinary metabolite profiles were significantly different between CD and UC patients. Previous NMR-based studies have shown that faecal metabolite profiles were more dysregulated in CD patients compared with UC patients.²⁷

4.2. Prognostic metabolic phenotypes are associated with primary response to anti-TNF therapy in Crohn's disease

A systematic difference was observed in the metabolic profiles of all three biofluids in moderate to severe CD patients who responded to the therapy over the course of treatment with anti-TNF drugs. Biomarkers that may contribute to the anti-TNF therapy non-response included serum lipids [phosphocholines, ceramide, sphingomyelin, and triglycerides], serum and faecal bile acids, histidine, and urinary cysteine. The combined profile of faecal lipids significantly differentiated CD anti-TNF responsive patients from non-responders, with a ROC curve for prediction of response demonstrating a sensitivity of 0.92 and specificity of 0.61. There are no strong predictors of response using clinical parameters such as FC or CRP, and our current profiles outperform these conventional biochemical markers.⁴⁴ A fall in FC after treatment predicts response to therapy.⁴⁵ Notably, the issues remain that the pre-treatment levels of CRP or FC do not act as predictors to anti-TNF therapy. It is the change in pre- and post-therapy FC that is associated with response.^{11,46} ROC curves created for serum and faecal bile acid profiles also yielded relatively strong predictive capacity, with sensitivity 0.92 and specificity of 0.61, respectively.

This study demonstrates that CD patients with higher levels of deoxycholic acid [secondary bile acids] during anti-TNF therapy were more likely to respond to anti-TNF therapy compared with CD patients with low levels. This was the same for some tertiary bile acids. Bile acids require further conjugation to change their

solubility to be eliminated in urine and faeces. CD patients who did not respond to anti-TNF therapy had higher levels of circulating primary unconjugated bile acids [cholic acid and chenodeoxycholic acid, Supplementary Figure S7], which may explain the significantly high level of bile acids conjugated to sulphate, taurine, and glycine in urine and faeces. Bile acids are an important pathway for inflammatory activation of the gastrointestinal tract.^{42,47} It has been demonstrated that primary bile acids are converted to secondary bile acids by some members of the gut microbiota, which underscores the potential role of the microbiome in the aetiology and progression of CD.

Lipid profiling revealed important biomarkers, with ceramide and sphingomyelin being elevated in primary non-responders in serum. Recent studies have demonstrated different inflammatory pathways, such as sphingosine-1-phosphate, which is known to be elevated in IBD patients.⁴¹ Sphingolipid metabolism and its downstream metabolites, such as ceramide and sphingomyelin, are known to be pro-apoptotic via dephosphorylation of Akt and activation of caspases, as well as being important in activation-induced cell death of dendritic cells.⁴⁸ Ceramide and sphingosine-1-phosphate are both pro-inflammatory metabolites, which promote egress of lymphocytes from lymph nodes. This pathway has been identified as a target for trials in both UC and CD.⁴⁹ The finding from our study, that sphingomyelin and ceramide were increased in non-responders, may represent a biomarker for patients whereby a pharmacodynamic reason is the cause for anti-TNF failure.

The bile acid diversity may be affected in our cohort of anti-TNF responders, as a result of alterations in microbiome occurring from CD activity and mucosal inflammation as seen between healthy controls and Crohn's patients. Previous literature has shown uncertainty as to whether gut dysbiosis leads to altered metabolism of bile acids or vice versa.⁵⁰ Dysbiosis within the small bowel lumen, with a tendency toward decreased bacteria-bearing bile salt hydrolase activity, may explain the increase in conjugated primary bile acids and a decrease in secondary bile acid pool.⁴² The findings have been demonstrated in the existing literature, with germ-free mouse experiments showing a loss of function in bile acid transformation pathway resulting in increases in primary, conjugated and sulphated bile acids.⁴²

The direct effects of bile acids may be their impact on mucosal barrier function.⁵⁰ However, some secondary bile acids may be protective and in fact decrease luminal inflammation via mechanisms

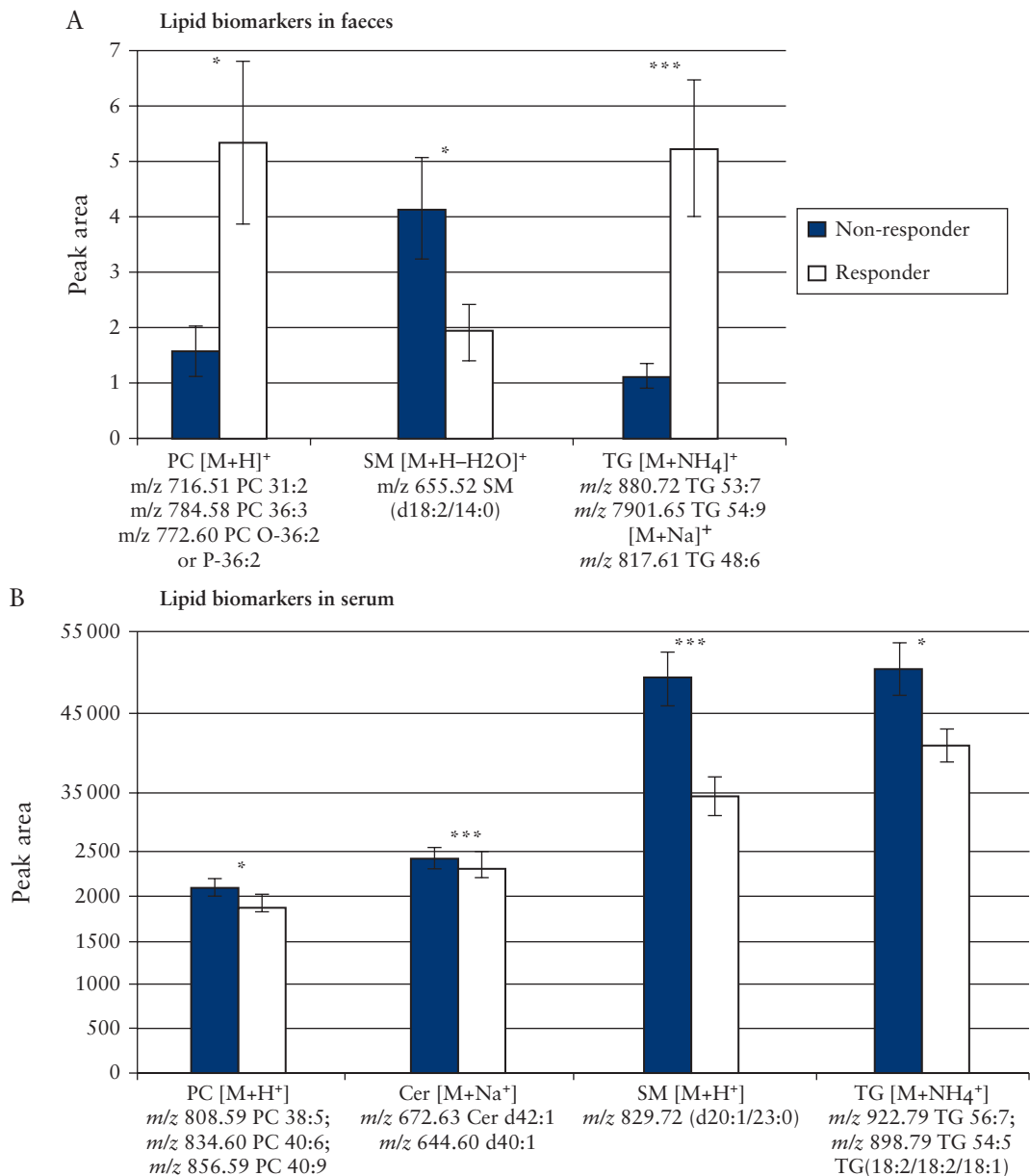


Figure 6. Lipid classes correlating with the longitudinal response of CD patients to anti-TNF therapy. Mean levels of lipid markers according to classes [PC, Cer, SM, and TG] which vary between patients responders and non-responders to anti-TNF treatment in faeces [A] and serum samples [B]. Two-tailed Student's t test, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. CD, Crohn's disease.

Table 2. Likelihood ratio for each AUC of lipid, bile acid and polar metabolite profiling.

	Faeces	Serum	Urine
Lipid profiling	2.36	2.25	-
Bile acids	2.36	2.36	2.36
Polar metabolite profiling	0.89	2.50	1.39

AUC, area under the curve.

outlined previously in the literature.⁵¹ Possible confounding due to the effects of antibiotics on numerous taxa was taken into account, as all patients did not have antibiotics within the last 6 months of their initial visit.

The profiling of polar metabolites showed low urinary cysteine as well as low faecal and serum histidine in non-responders to anti-TNF compared with responders, which may indicate that this pathway is of importance in anti-TNF response. There is emerging evidence that mucosal microbiome is involved in mammalian histamine production which alters mucosal pro-inflammatory cytokines. *Lactobacillus reuteri* may ameliorate mucosal inflammation through its impact on luminal histamine signalling via diacylglycerol kinase.⁵² In patients who did not respond to anti-TNF therapy, histidine metabolism might be disturbed as it is a precursor of histamine produced by mast cells that play a central role in the inflammatory response. Histidine is negatively correlated with CRP in CD, with decreased histidine levels being associated with risk of relapse in patients with UC.⁵³ In a murine *in vivo* model, histidine was shown to ameliorate colitis activity in dextran sulphate sodium [DSS]-colitic

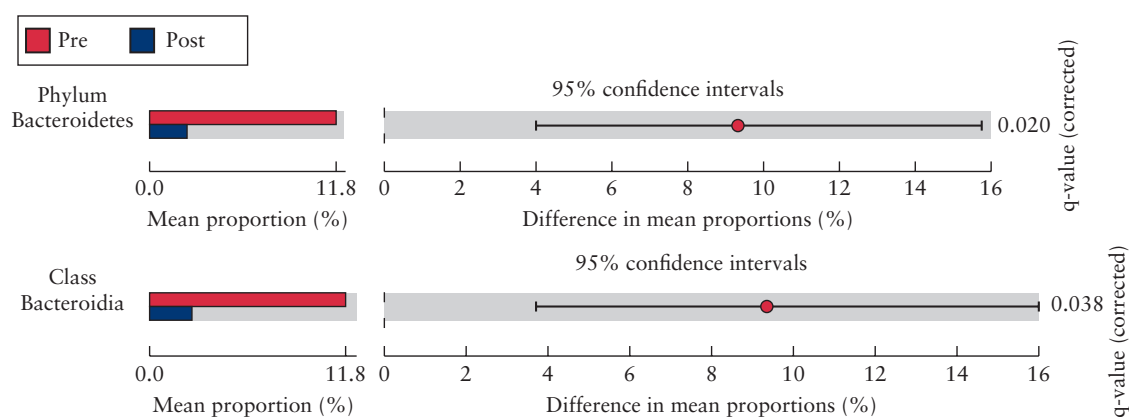


Figure 7. Differences in bacterial composition of IBD patients before and after receiving anti-TNF therapy. IBD patients included responders, non-responders, and intermediate responders, and post-samples were taken from one time point 5–7 months after treatment. IBD, inflammatory bowel disease.

mice by suppressing the production of pro-inflammatory cytokines, TNF- α , and IL-6, and inhibiting NF- κ by macrophages.⁵⁴

Biomarkers, including a panel of faecal, urine, and serum bile acids along with cysteine, have each shown an AUC of >0.7. This represents a novel panel of metabonomic-derived biomarkers predicting response to anti-TNF therapy in CD.

4.3. Metataxonomic

Anti-TNF therapy did not significantly alter the bacterial richness of the gut microbiota in IBD patients. In fact, it appeared as though there was relative stability in the microbial taxa. During anti-TNF treatment, *Bacteroidia* decreased when one time point following treatment was analysed [at baseline and one sample taken at 5–7 months]. There were no significant differences when samples were classified according to response status by using pairwise comparisons, likely owing to the small sample size. Stratification of IBD patients into responders vs non-responders to anti-TNF therapy, using the response index, showed trends in several bacterial taxa which did not reach statistical significance.

4.4. Limitations, current, and future work

Although this constitutes the largest longitudinal cohort study of IBD patients using metabonomic and metataxonomic profiling to date, the relatively small numbers of responders and non-responders from this cohort may explain the lack of statistical significance demonstrated by some assays and the 16S rRNA gene sequencing datasets. However, the small number of samples by responder class is overcome by the well-defined phenotype of Crohn's patients who have moderate to severe luminal disease, rather than a heterogeneous IBD cohort as with many of the previous published studies. Previous studies in the field have only recruited approximately 20 patients and similar numbers of controls in order to differentiate IBD from healthy or CD from UC, and mostly have not used Benjamin-Hochberg FDR.^{27,55} Furthermore, the longitudinal approach employed in this study allowed us to demonstrate alterations in metabolic and microbiota profiles, which have not been demonstrated or co-analysed in previous studies. Our study also endeavoured to be rigorous in defining patients who clearly responded to anti-TNF therapy vs those who did not respond, based on objective, in addition to subjective, markers of inflammation. Moreover, this study has provided possible mechanistic insights into causes of non-response to anti-TNF therapy through various pathways such as bile acid and fatty acids, which require further elaboration with *in vivo* models.

Metabonomics and metataxonomics may represent less invasive techniques than the only current predictive biomarker of long-term response to anti-TNF therapy: through endoscopy demonstrating mucosal healing.³⁸ Currently the analysis of samples is expensive and requires significant expertise, but with greater knowledge about which taxa or assays are important, targeted assays can be developed, allowing for faster and cheaper analysis to be performed.

In summary, we showed that metabonomic profiling uncovered candidate biomarkers associated with primary response to anti-TNF therapy in patients with Crohn's disease. These data demonstrate that anti-TNF therapy response may be associated with a patient's metabonomic and microbiota profiles, with important pathways of inflammation such as bile acid metabolism, lipid metabolism [PC, ceramide, sphingomyelin, triglyceride], and amino acid metabolism, all representing possible biomarkers of response to anti-TNF therapy in Crohn's disease. These findings outline a promising prognostic test for anti-TNF therapy response, and require validation in independent patient cohorts in IBD and other diseases that adopt anti-TNF therapy, such as rheumatoid arthritis.

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Conflict of Interest

None to declare.

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Author Contributions

ND, concept and design of the work, data collection, data analysis and interpretation, drafting of the article, critical revision of article, final approval of version to be published. MHS, data collection, data analysis and interpretation, drafting of the article, critical revision of article, final approval of version to be published. JAKM, data collection, data analysis and interpretation,

drafting of the article, critical revision of article, final approval of version to be published. APM, data analysis and interpretation. SOA, drafting of the article, critical revision of article, final approval of version to be published. RM, drafting of the article, critical revision of article, final approval of version to be published. PH, drafting of the article, critical revision of article, final approval of version to be published. LP, data collection, drafting of the article, critical revision of article, final approval of version to be published. EH, concept and design of the work, data analysis and interpretation, drafting of the article, critical revision of article, final approval of version to be published. JRM, concept and design of the work, data analysis and interpretation, critical revision of article, final approval of version to be published. ALH, concept and design of the work, data collection, data analysis and interpretation, drafting of the article, critical revision of article, final approval of version to be published.

Supplementary Data

Supplementary data are available at ECCO-JCC online.

Podcast

This article has an associated podcast which can be accessed at <https://academic.oup.com/ecco-jcc/pages/podcast>

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